

The arc is fine above 4 kHz. Below 2 kHz the room bends it by up to 8 dB.

For these runs the arc was laid flat (horizontal), with the propeller axis vertical. All microphones then sit at the same height, in the propeller plane, around the propeller. A perfect setup reads the same level at every position: a circle. We did this twelve times (keys prop7-18, 1-2 Sept 2026, PWM 1900). What is left after we remove each microphone's own offset is the effect of the room.

2026-09-08
data: laptop backup of 2 Sept
script: scripts/arc_error_map.py
full guide: docs/arc-validation-guide.html

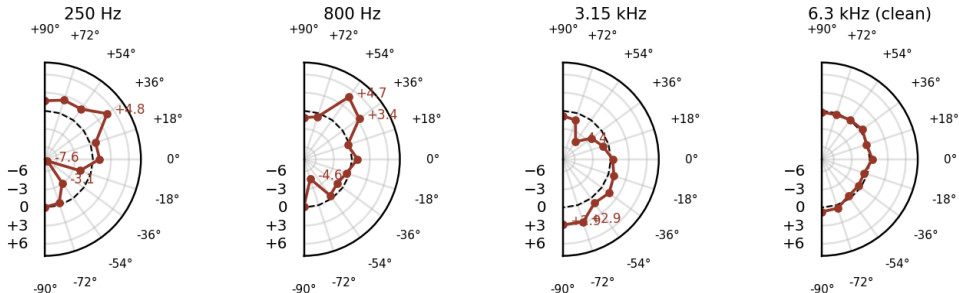


Fig. 1 · room error P(position), dB from a circle, mic offsets removed · fit over 12 captures: keys dp1-baseline-horizontal-prop7...prop18, PWM 1900 step (ids in the table below)

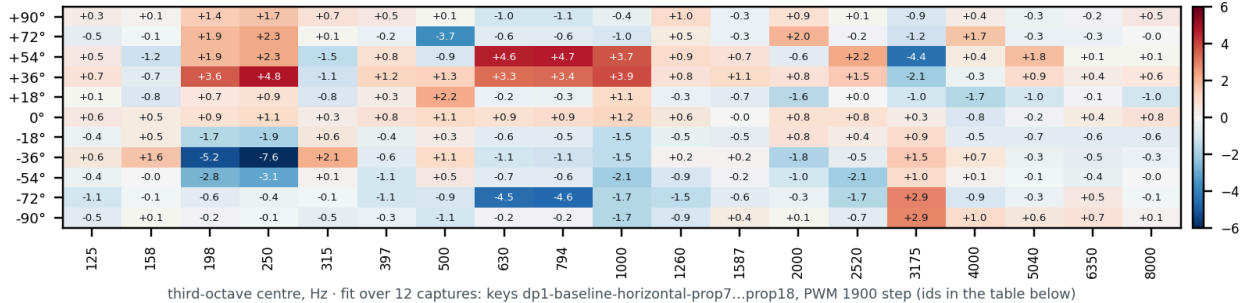
What the room does

- The level at -72° is 2.4 to 5.7 dB too low between 500 and 2000 Hz, in all twelve runs.** The dip is deepest at 630 and 800 Hz (-4.5 dB). It stays when a different microphone sits there (811-1896 in eight runs, 810-8901 in the four runs with the arc turned around). This position is the one closest to the chamber corner.
- The level at $+36^\circ$ and $+54^\circ$ is 3 to 7 dB too high, at 200-250 Hz and at 630-1000 Hz.** Three different microphones sat at $+54^\circ$ over the twelve runs. The bump stayed the same.
- Around the blade tone (200-250 Hz) the two halves of the arc disagree:** -36° reads -7.6 dB, $+36^\circ$ reads $+4.8$ dB; -54° reads -3 dB, $+54^\circ$ and $+72^\circ$ read $+2$ dB. The negative-angle half of the arc (-36° , -54°) is too low, the positive-angle half ($+36^\circ$, $+54^\circ$, $+72^\circ$) is too high. The dip also moves with motor speed (-9 dB at PWM 1900, -2 dB at 2000). A sound wave that bounces off a surface and adds to the direct wave does exactly this: it cancels at some frequencies and adds at others, and the tone slides through those frequencies as the speed changes.
- Above 4 kHz the arc reads a circle within ± 1 dB at every position.** The positions 0° , $\pm 18^\circ$ and $\pm 90^\circ$ are within about 1 dB in every band.

What it is not

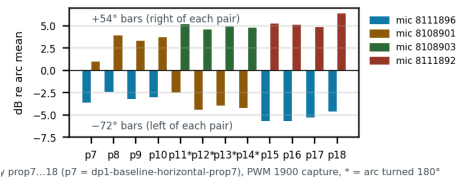
- Not the microphones**, with one exception. Because the microphones at $+72^\circ$ and $+54^\circ$ were swapped after prop10, and the arc was turned around for prop11-14, we can tell apart what belongs to a microphone and what belongs to a position. Ten microphones agree within ± 1.8 dB in every band. **Microphone 811-1892 reads 2-6 dB too low from 125 Hz to 2.5 kHz.** Its calibration file does not match the capsule (its sensitivity value, -10.63 dB, is the least negative of the set, while the raw capsule is only about 1 dB low). When it sat at $+54^\circ$ it made the room error look like 7 dB. Take it off the arc or measure it again.
- Not a labelling error.** prop11, 12, 13 and 14 were measured with the arc turned around. Their patterns match the mirror image of the other runs (correlation $+0.79$ to $+0.98$). Plotted by label, the dip appears on the other side. Plotted by physical position, they agree with the other eight runs.
- Not one frequency.** The error sits in three separate bands, 200-250 Hz, 630-1000 Hz and 3150 Hz, and nowhere else (Fig. 2). Rajmane and Baumann (DAGA 2016) put steel plates into a good chamber and found that a plate only shows up once the wavelength is shorter than the plate: a 35 cm plate from 1.25 kHz, a 10 cm plate from 3.15 kHz. So three bands point to three objects of different size: about 1.5 m (floor, table or wall), about 0.4 m (stand base or a bar of the arc frame), about 0.1 m (a microphone clamp or mount).

Fig. 2 · Room error at each position and frequency band, in dB from a circle (microphone offsets removed; run-to-run scatter 0.6-2.3 dB)



third-octave centre, Hz · fit over 12 captures: keys dp1-baseline-horizontal-prop7...prop18, PWM 1900 step (ids in the table below)

Fig. 3 · Two positions, twelve runs, three microphones



500-2000 Hz, microphone offsets removed; * = arc turned around. The dip at -72° and the bump at $+54^\circ$ keep their size no matter which microphone sits there.

Measurements plotted (all figures): the last PWM-1900 capture in each key 2004_6in_unset_dp1-baseline-horizontal- <key>, 11 microphones $\times$ 2.0 s, calibrated; microphone serials 810-/811-xxxx given by their last four digits

KEY	CAPTURE (PWM 1900)	ARC	+72/+54	KEY	CAPTURE (PWM 1900)	ARC	+72/+54	KEY	CAPTURE (PWM 1900)	ARC	+72/+54
prop7	2026-09-01 16:42:31	norm	1892/8901	prop11	2026-09-02 13:18:49	180°	8901/1892	prop15	2026-09-02 15:48:59	norm	8901/1892
prop8	2026-09-01 16:44:39	norm	1892/8901	prop12	2026-09-02 14:24:12	180°	8901/1892	prop16	2026-09-02 15:54:26	norm	8901/1892
prop9	2026-09-01 16:50:45	norm	1892/8901	prop13	2026-09-02 14:34:37	180°	8901/1892	prop17	2026-09-02 16:52:44	norm	8901/1892
prop10	2026-09-01 16:55:35	norm	1892/8901	prop14	2026-09-02 14:37:02	180°	8901/1892	prop18	2026-09-02 17:08:19	norm	8901/1892

Left out of every figure: the 31 Aug keys (...-2026-08-31, -2...-6) and prop1-prop6 (1 Sept morning). Their patterns match neither arc orientation, so the rig or the room was different then. "180°" means the label +e was physically at -e; those four runs are mirrored before plotting. The offset of microphone 811-1892 is removed everywhere. Figs. 1-2 show the fitted position term P; Fig. 3 shows single runs.

Method: Welch spectrum (4096-point Hann), UMIK-2 calibration as the app applies it, third-octave sums from 125 Hz to 8 kHz. Model: level = run gain + position term P + microphone term M, fitted by least squares over prop7-18 with prop11-14 mirrored. Every number comes from the files named above; CSV tables in docs/analysis/. Literature: papers/small-chamber/ and papers/chamber-problems/.